

Problem Set 10

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MinHash

Problem 1. The matrix below has six rows (elements 0–5) and four columns, where each column is the characteristic vector of a set S_j (a 1 in row i means element $i \in S_j$).

<i>Element</i>	S_1	S_2	S_3	S_4
0	0	1	0	1
1	0	1	0	0
2	1	0	0	1
3	0	0	1	0
4	0	0	1	1
5	1	0	0	0

(a) Compute the minhash signature for each column if we use the following three hash functions:

$$h_1(x) = 2x + 1 \bmod 6, \quad h_2(x) = 3x + 2 \bmod 6, \quad h_3(x) = 5x + 2 \bmod 6.$$

(b) Which of these hash functions are true permutations (of the rows $\{0, 1, \dots, 5\}$)?

(c) How close are the estimated Jaccard similarities for the six pairs of columns to the true Jaccard similarities?

Concentration

Problem 2: Expectation, Variance, and Chebyshev.

Let X be a discrete random variable with the following probability mass function:

x	0	1	2	3	4	5	6	7
$\Pr[X = x]$	0.05	0.10	0.15	0.20	0.20	0.15	0.10	0.05

(a) Compute the expected value $\mathbb{E}[X]$.

(b) Compute the variance $\text{Var}[X]$. *Hint:* $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

- (c) Use **Chebyshev's inequality** to give an upper bound on $\Pr[|X - \mathbb{E}[X]| \geq 3]$. Then compute this probability *exactly* from the table. Is Chebyshev's bound tight here?
- (d) Now let $X_1, \dots, X_n$ be independent copies of X , and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be their average. State $\mathbb{E}[\bar{X}_n]$ and $\text{Var}[\bar{X}_n]$.
- (e) Using Chebyshev's inequality, how large must n be to guarantee

$$\Pr[|\bar{X}_n - \mathbb{E}[X]| \geq 0.1] \leq 0.05?$$

Report a concrete number.